

Don't write
wrong spelling

always end with ;

Page No.

Date

Commands :-

run - control + enter
with hashtag . Comments (Notes)

Types of commands.

DDL, DQL, DML, TCL, DCL

DDL - DATA DEFINITION LANGUAGE

1 - CREATE

CREATE DATABASE 10-30-Batch ;

Use the database - 1st line

Use 10-30-BATCH ;

CREATE TABLE employees (
emp-id INT PRIMARY KEY ,
emp-name VARCHAR(50)) ;

2 - ALTER - CHANGE THE STRUCTURE

OF ALREADY CREATED DATABASE /
TABLE

ALTER TABLE employees ~~TO~~ emp
ADD email VARCHAR(100) ;

3 RENAME -

RENAME TABLE employees to emp-details;

(DROP, DELETE AND TRUNCATE)

4 TRUNCATE - delete all the data only
TRUNCATE TABLE emp-details;

5 DROP - DELETE PERMANENTLY
DROP TABLE emp-details;

DROP DATABASE io_30 batch;

DDL - DATA DEFINITION
LANGUAGE

CREATE, RENAME, ALTER, DROP,
TRUNCATE

2 - DML - DATA MANIPULATION
LANGUAGE

INSERT, UPDATE, DELETE

Formula $\rightarrow$ command -

* CREATE DATABASE io BATCH;

* USE io BATCH;

CREATE TABLE student (student-id IN
PRIMARY KEY, name VARCHAR(50));

INSERT

Formula $\rightarrow$ command

$\rightarrow$ INSERT INTO student (student_id , name
VALUES (01, "sagar") , (02, "yash");

$\rightarrow$ INSERT INTO student (student_id , name)
VALUES (03, "Nita");

For watching table use this :-

Select * From student ;
 $\uparrow$ $\uparrow$
Star Table Name

$\rightarrow$ UPDATE student
SET student_id = 05
Where name = "Nita" ;

****** SET SQL_SAFE_UPDATES = 0 ;

$\rightarrow$ Use for open safe mode.

for off safe mode use this

SET SQL_SAFE_UPDATES = 1 ;

→ DELETE - PARTICULAR RECORD

```
DELETE FROM student
WHERE student_id = 05;
```

3 - DQL = Data Query Language

```
SELECT columns name from table
name
```

* star use for all columns

```
SELECT student_id FROM student;
```

```
SELECT name FROM student;
```

```
SELECT student_id, name FROM student;
```

```
SELECT * FROM student;
```

TCL - TRANSACTION CONTROL LANGUAGE

- 1) COMMIT - save
- 2) ROLLBACK - UNDO
- 3) SAVEPOINT - जितना दुबारा 3000 सेव करो
- 4) ROLLBACK TO SAVEPOINT

DCL - DATA CONTROL LANGUAGE

1. GRANT - GIVE THE PERMISSION
2. REVOKE - CANCEL THE PERMISSION

DDL, DCL, TCL, DQL, DML

DATA TYPE

NUMERICAL

INT - INTEGER = 100, 500, 454

TINYINT - SMALL INTEGER - 1, 2

DECIMAL (10, 2) - 5000.45

FLOAT

DOUBLE - LARGER FLOAT NO.

BIGINT

CATEGORICAL

CHAR - CHARACTER - fixed length text

~~VARCHAR~~ VARCHAR (variable character)

VARCHAR (100) - TUSHAR

DATETIME - 2026-9-8 11:35:20

YEAR - 2026

MONTH - 9

DAY - 8

* Constraints -

→ constraints are rule/regulations applied to column in table to control what kind of data can be stored.

1) PRIMARY KEY - It's a unique key
→ Roll No, chapter no, id, sr. No.

2) NOT NULL
→ mo. No., emailid

3) UNIQUE
→ Email ID.

4) DEFAULT
→ PINCODE - WARDANA - 442001

5) CHECK
→ CHECK AGE > 18
GENDER = "FEMALE"
AGE > 60

6) FOREIGN KEY
→ PRIMARY KEY OF THE OTHER TABLE

7) AUTO INCREMENT

→ CREATE DATABASE company ;
USE company ;

CREATE TABLE departments (dept-id INT
PRIMARY KEY AUTO-INCREMENT,
dept_name VARCHAR(50) NOT NULL UNIQUE);

~~INSERT INTO departments ;~~

~~CREATE TABLE employees (emp-id INT)~~

INSERT INTO departments
(dept-name) VALUES ("IT"), ("HR"), ("FINANCE"),
("MARKETING"), ("SALES");

CHECKING the table

SELECT * FROM departments.

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 CREATE TABLE emp employees (emp-id INT
PRIMARY KEY AUTO-INCREMENT, emp-name
VARCHAR(50) NOT NULL, email VARCHAR(100)
UNIQUE, age INT CHECK (age >= 18),
salary DECIMAL(10,2) CHECK (salary > 0),
status VARCHAR(20) DEFAULT 'ACTIVE',
dept-id INT, FOREIGN KEY (dept-id) REFERENCES
departments (dept-id));

INSERT INTO employees
(emp-name, email, age, salary, status, dept-id)
VALUES
("Sudhiksha", "Sudhiksha@gmail.com", 25, 45000,
DEFAULT, 1), ("Nikita", "Nikita@gmail.com", 26,
67000, "NON-ACTIVE", 2);

INSERT INTO employees
(emp-name, email, age, salary, status, dept-id)
VALUES
("Prathamesh", "Pratham@gmail.com", 19, 1, DEFAULT,
1);

Error :-

Code → 1064 → syntax error
↓
Code structure

always ending with the semicolon (;)

ALTER → वस्तुके से change करना

TRUNCATE → खाली करना

DROP → Delete permanently

code → 1175

जब भी 3-बार है तो

safe mode on हो जाता है

savepoint → जिला हुआ जगह से
लेव करो

1046 → error for not selecting the database

first select the database

offset → छोड़के